NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE
(AUTONOMOUS)
PAMPADY, THIRUVILWAMALA, THRISSUR DT., KERALA - 680588
NBA ACCREDITED PROGRAMMES - B.TECH CSE, B.TECH MTR | NAAC RE-ACCREDITED WITH 'A' GRADE
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Since 1968

DEPARTMENT OF MCA
QUESTION BANK

SUBJECT :24MCACNT202 ADVANCED COMPUTER NETWORKS
SEM/YEAR: II/I
Subject Handled By: Sumi M & Aparna A

Module I

Overview of Computer Networks – Types, Topologies; OSI model, TCP/IP, Concept of Quality of Service (QoS), Client-server &Peer-to-peer architecture, Application layer protocols- HTTP, FTP, SMTP, POP3, and DNS

Q No	Question	BTL	Competence
1.	Explain Computer Network	2	Understand
2.	A network must be able to meet certain criteria's. Explain	1	Remember
3.	Explain Basic Communication Model with neat Diagram	1	Remember
4.	Explain the types of Computer Networks	2	Understand
5.	Explain the purpose of each OSI layers in detail.	2	Understand
6.	Explain OSI reference model in detail	1	Remember
7.	Define Quality of Service in detail	2	Understand
8.	Describe the Techniques to Improve QoS	1	Remember
9.	Explain the following: a) Client-server architecture b) Peer-to-peer architecture	2	Understand
10.	Explain the Working of DNS	2	Understand
11.	Write a short note on SMTP	2	Understand
12.	Describe POP3 and DNS	2	Understand
13.	Differentiate HTTP persistent and non-persistent communications.	2	Understand
14.	Explain the two predominant network architecture used in modern network applications with diagram	2	Understand
15.	Explain the working of File Transfer Protocol and its features	2	Understand
16.	Differentiate OSI model and TCP/IP model with examples.	2	Understand

Module II

Introduction to Transport layer protocols: Services, Protocols- Connection oriented transport TCP, Connection less transport UDP, Multiplexing , Principles of Reliable data transfer - Stop-and-wait and Go-back- N

Q No	Question	BTL	Competence
1.	What are the Services provided by the Transport Layer	2	Understand
2.	Define Stop and Wait Protocol with neat diagram	2	Understand
3.	Explain User Datagram Protocol	2	Understand
4.	Explain the Working of Go-Back-N ARQ with neat diagram	1	Remember
5.	Explain the working of TCP and UDP models in modern computing. How it helps for reliable data transfer.	3	Application
6.	Explain TCP Segment Format	2	Understand
7.	Define Stop and Wait Protocol with neat diagram	2	Understand
8.	Explain the protocols in Transport Layer	2	Understand
9.	Explain the Principles of Reliable Data Transfer	2	Understand
10.	Explain Frequency-division Multiplexing (FDM)	3	Application
11.	What are there are two types of TDM. Explain	2	Understand
12.	Define Time Division Multiplexing	2	Understand
13.	Wavelength Division Multiplexing (WDM)	1	Remember
14.	Explain the functions of each networking devices in computer networking.	3	Application
15.	Differentiate rdt1.0 and rdt2.0 with neat diagram	1	Remember
16.	Differentiate Connection oriented transport Protocol and Connection less transport Protocol	1	Remember
17.	Differentiate the working of multiplexing and de-multiplexing with neat diagram.	2	Understand
18.	Explain the practical applications of principles of reliable data transfer in computer networking.		

Module III

Virtual circuits and datagrams. Principles of routing. Internet protocol Ipv4 CIDR. Routing algorithms: Link-state and distance vector routing. Routing on the internet RIP, OSPF and BGP, Multicast routing. Introduction to IPV6

Q No	Question	BTL	Competence
1.	Differentiate virtual circuit and Datagrams	2	Understand
2.	Describe the steps involved in the link-state routing protocol, including the creation and dissemination of link state advertisements (LSAs), and the calculation of the shortest path using an example.	2	Understand
3.	Compare and contrast two interior gateway protocols used in the internet, highlighting their characteristics, advantages, and disadvantages	1	Remember
4.	Explain Routing information protocol (RIP)	2	Understand
5.	Differentiate TCP and UDP protocols with one example each.	2	Understand
6.	Explain the role of inter domain routing protocol in detail	2	Understand
7.	Explain about routing and its types	2	Understand
8.	Differentiate adaptive and non-adaptive algorithms.	2	Understand
9.	Describe the border gateway protocol (BGP) and its role in inter-domain routing, including its basic components, message types, and the routing calculation process	2	Understand
10.	Define path attributes and its categories.	2	Understand
11.	Describe the OSPF protocol, including type of packets, links and its basic components	2	Understand
12.	Define autonomous system and its types.	2	Understand
13.	Define link state algorithm with the help of example.	2	Understand
14.	Explain a) Classfull Addressing b) CIDR	2	Understand
15.	Explain the concept of routing algorithm and describe the different types of routing algorithms used in computer networks.	2	Understand
16.	Explain the role of intra domain routing protocol in detail	2	Understand
17.	Explain how the emergence of CIDR became the solution for classfull addressing.	3	Application

Module IV

Introduction to link layer - Error detection (parity, checksum, and CRC). Multiple access protocols (collision and token based). IEEE 802.3 Ethernet, Switching and bridging, Data encoding. Ethernet switches.

Q No	Question	BTL	Competence
1.	What the main purpose of CRC generator and CRC checker for error detection in link layer?	2	Understand
2.	Explain briefly on error detection code technique checksum used in data communication. For this given data 11001100 10101010 11110000 11000011, perform check sum operation at sender site and receiver site and verify the data at receiver site.	2	Understand
3.	Explain Carrier Sense Multiple Access with collision detection algorithm in detail	3	Application
4.	Define MAC address and its types.	2	Understand
5.	Define CRC. With the help of an example explain the working of CRC generator and CRC checker for error detection.	3	Application
6.	Define channelization protocol and its types.	1	Remember
7.	Explain about data link layer and the services provided by it	2	Understand
8.	Explain about error detection techniques in link layer.	2	Understand
9.	Compare and contrast three multiple access control protocols in terms of Performance, advantages and disadvantages. How do these protocols handle collisions, and what are the implications for network efficiency and reliability?	2	Understand
10.	Explain in detail on a) switching b) bridging	3	Application
11.	Explain about digital encoding and its categories with examples.	3	Application
12.	Define multiple access protocols. How it is classified?	3	Application
13.	Define switching and its types in detail.	3	Application
14.	Explain about random access protocols and its categories in detail.	2	Understand
15.	Explain: a) Single parity check b) Two dimensional parity check	1	Remember
16.	Differentiate single parity check and 2 dimensional parity check used for error detection in data link layer.	2	Understand

Module V

IEEE 802.11 Wi-Fi, Bluetooth, Threats and attacks. Network Address Translation. Firewalls, VPN,SNMP.

Q No	Question	BTL	Competence
1.	Describe the IEEE 802.11 Architecture in detail	2	Understand
2.	Illustrate Frame Format of IEEE 802.11	2	Understand
3.	What are the advantages and disadvantages of IEEE 802.11	1	Remember
4.	The architecture of Bluetooth defines two types of networks. Explain	2	Understand
5.	Explain Bluetooth protocol stack with neat diagram	2	Understand
6.	Explain • Types and applications of Bluetooth • Advantages and disadvantages of Bluetooth	1	Remember
7.	State and explain types of network attacks	1	Remember
8.	What are the Common Types of Network Attacks? Explain	2	Understand
9.	Explain the Working of Network Address Translation	1	Remember
10.	What are the 3 ways to configure NAT. Explain	2	Understand
11.	Explain Simple Network Management Protocol	2	Understand
12.	Define Firewall and its types	1	Remember
13.	What are the Types of Virtual Private Network (VPN) Protocols? Explain	1	Remember
14.	Describe the five types of messages in SNMP	2	Understand
15.	Write a short note on virtual private network	1	Remember
16.	Explain the Advantages and Disadvantages of Firewall	1	Remember
17.	Explain the different kinds of network security mechanisms in networking.	2	Understand
18.	Explain network threats and attacks in detail.	2	Understand
19.	Explain the role of SNMP in network management.	2	Understand